



1. What is the value of

$$(2 + 0 + 2 + 1)^2 - (2 - 0 + 2 - 1)(2 + 0 - 2 + 1)?$$

(A) 0 (B) 13 (C) 22 (D) 23 (E) 25

2. What is $\gcd(2020 \cdot 2021, 2019 \cdot 2022)$?

(A) 1 (B) 2 (C) 3 (D) 6 (E) 8

3. Suppose m, n are positive integers such that $m + 1, m + 2, \dots, m + n$ are all semiprimes (product of 2 distinct primes). What is the largest possible value of n ?

(A) 2 (B) 3 (C) 4 (D) 5 (E) 6

4. What is the units digit of $2021^2 - 2^{2021}$?

(A) 1 (B) 3 (C) 5 (D) 7 (E) 9

5. How many nonsimilar triangles exist which have at least two lines of symmetry?

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

6. The difference between the maximum and the minimum 4-digit positive integer written using each of the digits 1, 2, 3, n exactly once is 2088. What is the sum of all the possible values of n ?

(A) 1 (B) 3 (C) 4 (D) 9 (E) 10

7. How many positive integers $n < 10^9$ written using only the digits 0 and 1 are divisible by 12?

(A) 40 (B) 42 (C) 47 (D) 53 (E) 56

8. If x, y, z are positive integers such that

$$x \cdot y \cdot z = 7!,$$

what is the smallest possible value of $\max(x, y, z)$?

(A) 14 (B) 16 (C) 18 (D) 20 (E) 21

9. How many ways can the letters of the word *COMBO* be arranged such that the M is adjacent to both a vowel and a consonant?

(A) 6 (B) 12 (C) 18 (D) 24 (E) 32

10. An equilateral triangle is inscribed in another equilateral triangle as in the diagram below. If the side length of the outer equilateral triangle is 1, the side length of the inner triangle can be represented as $\frac{\sqrt{a}-\sqrt{b}}{c}$ for positive integers a, b , and c such that a and b are square free. What is $a + b + c$?

(A) 10 (B) 12 (C) 13 (D) 15 (E) 16

11. Nine friends play a game. The first person says 1. For $1 \leq n \leq 8$, if the n th person says k then the $(n + 1)$ th person will say either $k - 1$ or $k + 1$ with equal probability. What is the probability that the number the 9th person says is positive?

(A) $\frac{163}{256}$ (B) $\frac{41}{64}$ (C) $\frac{165}{256}$ (D) $\frac{83}{128}$ (E) $\frac{167}{256}$

12. On the 31st December 2020, Andre realizes that the number of days until his 21st birthday is the square of the number of entire months leading to it. When does his birthday fall?

(A) May 16th (B) Jun 17th (C) Jul 18th (D) Aug 19th (E) Sep 20th

13. Rectangle $ABCD$ has $AB = 2$, $BC = 1$. Point P is selected uniformly at random inside $ABCD$. Through P are lines ℓ and m perpendicular to $\overline{AC}$ and $\overline{BD}$ respectively. What is the probability that both ℓ and m intersect segments AB and CD ?

(A) $\frac{3}{16}$ (B) $\frac{5}{16}$ (C) $\frac{3}{8}$ (D) $\frac{1}{2}$ (E) $\frac{5}{8}$

14. Angela is calculating her grades in her Calculus class by computing the total number of points she scored divided by the total number of possible points. On her last four tests, she scored 20 out of 30 points, 35 out of 45 points, 50 out of 60 points, and 65 out of 75 points. Suppose she scores exactly 98% of the possible m points and exactly 96% of the possible n points on her final two lengthy exams such that her final grade in the class is precisely 90%. Given that m , n , and the number of points she scored per test are positive integers, what is the smallest possible value of $m + n$?

(A) 175 (B) 200 (C) 225 (D) 250 (E) 275

15. Suppose n is a composite positive integer, and let $f(n)$ be the third-largest divisor of n . For how many positive integers $n \leq 720$ is $f(n)$ divisible by 4?

(A) 45 (B) 60 (C) 72 (D) 90 (E) 120

16. Let $ABCD$ be a parallelogram such that $AB = CD = 1$, $BC = DA = \sqrt{3}$, and $\angle BAD = \angle DCB = 30^\circ$. Let E , F , G , and H be the points such that $\triangle ABE$, $\triangle BCF$, $\triangle CDG$, and $\triangle DAH$ are equilateral triangles lying completely outside of $ABCD$. Compute the area of quadrilateral $EFGH$.

(A) $2\sqrt{6}$ (B) 5 (C) $3\sqrt{3}$ (D) $2\sqrt{7}$ (E) 6

17. Jensen has polynomials $(x-1)$, $(x-1)(x-2)$, $\dots$, $(x-1)(x-2)\cdots(x-2020)(x-2021)$ and he decides to split them in two groups. Let $P(x)$ and $Q(x)$ denote the product of all the polynomials in the first and in the second group, respectively. Given that $Q(x)$ is divisible by $P(x)$, what is the smallest possible degree of $\frac{Q(x)}{P(x)}$?

(A) 1 (B) 3 (C) 1009 (D) 1011 (E) 2021

18. For a positive integer n , let $f(n)$ be the product of the nonzero digits of n . Find the least n so that $f(1) + f(2) + f(3) + \dots + f(n) \geq 2021$.

- (A) 89 (B) 96 (C) 97 (D) 98 (E) 99

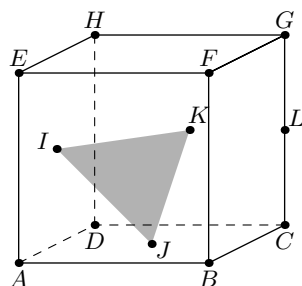
19. Let $ABCD$ be an isosceles trapezoid with bases AB and CD , such that $AB < CD$. Let ℓ be the altitude from B to AC . Suppose that $AD = BC = 5$, $AB = 4$, and ℓ bisects the area of $ABCD$. Compute CD .

- (A) $3\sqrt{5}$ (B) $4\sqrt{3}$ (C) 7 (D) $5\sqrt{2}$ (E) $2\sqrt{13}$

20. Four kids—Hyde, Heli, Lithe, and Beryl—request at least one, two, three, and four pieces of candy, respectively. However, they only have five (indistinguishable) pieces of candy to share. How many ways can the kids split up the candy such that at least one request is fulfilled?

- (A) 47 (B) 48 (C) 50 (D) 53 (E) 54

21. In the figure below, $ABCDEFGH$ is a cube with side length 10. If I is the center of $ADHE$, J is the center of $ABCD$, K is the center of $CDHG$, and L is the midpoint of $\overline{CG}$, then the distance between L and the plane formed by $\triangle IJK$ can be written as $\frac{m}{\sqrt{n}}$ for positive integers m, n such that n is square-free. Find $m + n$.



- (A) 7 (B) 8 (C) 11 (D) 12 (E) 13

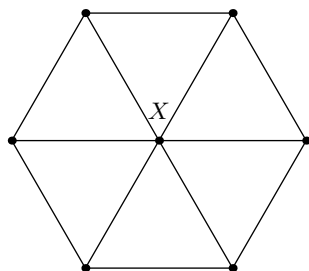
22. Define a positive integer $k \geq 2$ to be artistic if there exists a natural number m such that for any positive integer n that is relatively prime to k , we have

$$m^n + n^m \equiv 0 \pmod{k}.$$

Determine the sum of all artistic numbers.

(A) 2 (B) 5 (C) 8 (D) 13 (E) 18

23. An ant starts on point X in the following diagram. If the ant moves to an adjacent vertex every second, how many ways are there for the ant to return to point X after 6 seconds?



(A) 124 (B) 288 (C) 456 (D) 648 (E) 744

24. Triangle $\triangle PQR$ with $PQ = 3$, $QR = 4$, $RP = 5$ is drawn inside a regular hexagon $ABCDEF$ with P on segment FA , Q the midpoint of segment AB , and R on segment CD . Given that $AB^2 = \frac{m}{n}$ for relatively prime positive integers m and n , find $m + n$.

(A) 865 (B) 866 (C) 867 (D) 868 (E) 869

25. Let $p_1 < p_2 < \dots < p_{24}$ be the distinct permutations of 1234 in increasing order. For positive integers $1 \leq i < j \leq 24$, we define $d_{(i,j)} = p_j - p_i$. For example, $d_{(1,24)} = 4321 - 1234 = 3087$. Over all possible choices of (i, j) , how many distinct values of $d_{(i,j)}$ are there?

(A) 92 (B) 94 (C) 96 (D) 98 (E) 100

Time limit: 75 minutes